

Question:

Tom's age is \$T\$ years, which is also the sum of the ages of his three children. His age \$N\$ years ago was twice the sum of their ages then. What is \$T/N\$?

$\textbf{(A) } 2 \quad \textbf{(B) } 3 \quad \textbf{(C) } 4 \quad \textbf{(D) } 5 \quad \textbf{(E) } 6$

Reasoning + Answer:

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1. Understand the given information:

- Tom's current age is T years.
- The sum of the ages of his three children is also T .
- N years ago, Tom's age was twice the sum of the ages of his children at that time.

2. Define the ages of the children:

- Let the ages of the three children be a , b , and c .
- Therefore, $a + b + c = T$.

3. Determine Tom's age N years ago:

- Tom's age N years ago was $T - N$.

4. Determine the ages of the children N years ago:

- Each child's age N years ago would be $a - N$, $b - N$, and $c - N$.

5. Sum of the ages of the children N years ago:

- The sum of their ages N years ago is $(a - N) + (b - N) + (c - N)$.

6. Simplify the sum:

- $(a - N) + (b - N) + (c - N) = (a + b + c) - 3N$.

7. Substitute the given information:

- We know $a + b + c = T$.
- Therefore, the sum of the children's ages N years ago is $T - 3N$.

8. Set up the equation based on the problem statement:

- Tom's age N years ago was twice the sum of the children's ages N years ago.
- Therefore, $T - N = 2(T - 3N)$.

9. Solve the equation:

- $T - N = 2(T - 3N)$
- $T - N = 2T - 6N$
- $T - 2T = -6N + N$
- $-T = -5N$
- $T = 5N$

10. Find the ratio $\frac{T}{N}$:

- $\frac{T}{N} = \frac{5N}{N} = 5$.

11. Conclusion:

- The ratio $\frac{T}{N}$ is 5.

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<answer> 5 </answer>